YEAR 8 MATHS · CHAPTER 7 · MIND MAP

Equations and inequalities

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1 Reviewing equations

An equation has an equals sign. The solution is the value that makes it true.

$x + 7 = 12 \Rightarrow x = 5$ $3x = 21 \Rightarrow x = 7$

Check by substituting: $5 + 7 = 12$ ✓

2 Equivalent equations

Do the same to both sides. Undo in reverse order.

$2x + 3 = 11 \Rightarrow 2x = 8 \Rightarrow x = 4$

3 Equations with fractions

Clear the fraction by multiplying both sides.

$\frac{x}{3} = 5 \Rightarrow x = 15$

$\frac{x+1}{2} = 4 \Rightarrow x + 1 = 8 \Rightarrow x = 7$

$\frac{x}{4} + 1 = 3 \Rightarrow \frac{x}{4} = 2 \Rightarrow x = 8$

4 Pronumeral on both sides

Subtract the smaller term so the pronumeral is on one side.

$5x + 2 = 3x + 10 \Rightarrow 2x = 8 \Rightarrow x = 4$

5 Equations with brackets

Expand the brackets first.

$3(x - 2) = 12 \Rightarrow 3x - 6 = 12 \Rightarrow x = 6$

$2(x + 1) = x + 6 \Rightarrow 2x + 2 = x + 6 \Rightarrow x = 4$

6 Simple quadratic equations

Get x^2 by itself, then take the square root. Remember both answers.

$x^2 = 25 \Rightarrow x = \pm 5$

$x^2 + 3 = 19 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$

$x^2 = -4$ has no solution: a square can't be negative.

7 Formulas and relationships

Substitute what you know, then solve for what's left.

$P = 2l + 2w$ with $l = 5,\ w = 3$: $P = 16$

$A = lw$ with $A = 24,\ w = 4$: $l = 6$

8 Applications of equations

Let a letter be the unknown, write an equation, solve it, answer in words.

Double a number and add 5 to get 17: $2x + 5 = 17$, so the number is 6

Rectangle, width w, length $2w$, perimeter 30: $6w = 30$, so 5 by 10

9 Inequalities

$<$ less than, $\leq$ less than or equal, $>$ greater than, $\geq$ greater than or equal.

$x > 2$: open circle at 2, arrow right

$x \leq 4$: closed dot at 4, arrow left

10 Solving inequalities

Solve like an equation, but flip the sign when you multiply or divide by a negative.

$2x + 1 < 9 \Rightarrow 2x < 8 \Rightarrow x < 4$

$-3x \leq 12 \Rightarrow x \geq -4$

Watch out: flip the sign when dividing by a negative.